Student Seminar - IDP Presentation 7

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Updated Peptide Sequence Definition

Old: **Peptide sequence in letters. (e.g. 'FF')**

New: **The IUPAC amino acid sequence of the peptide using standard single-letter codes (e.g., 'ACDEFGHIKLMNPQRSTVWY'). Must contain only valid uppercase IUPAC single-letter residues. Do not include spaces, hyphens, numbers.**

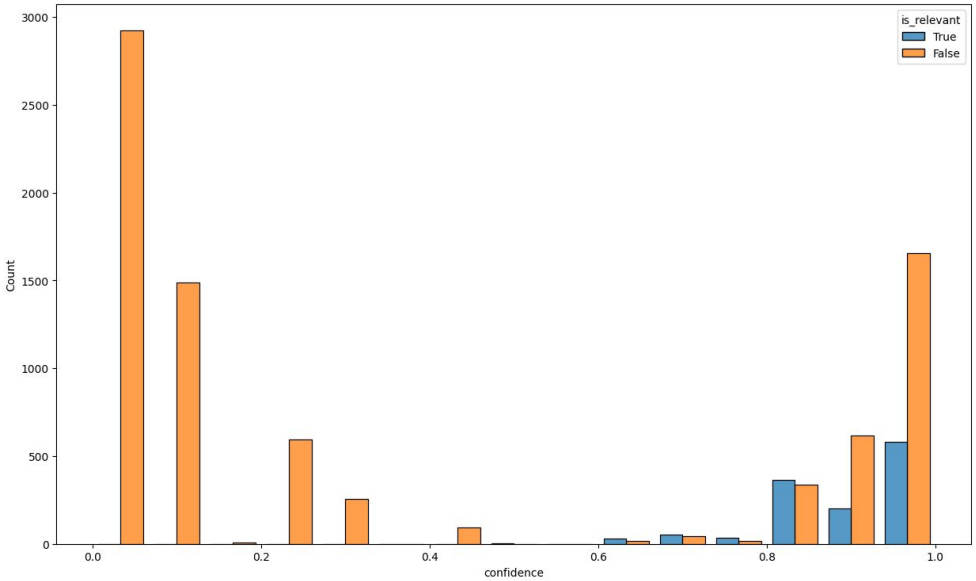
100 paper sample:

Before: 37/228 sequences valid

Now: 99/160 sequences valid

- Sequence count dropped but valid extraction rate increased
- There is an edge case (e.g. DIPHENYLALANINE)

Relevance Check



Gemini 3.5 flash lite

250 papers sampled (different decisions on relevancy) extractions

- Gemini 3.5 Flash Lite: 158 correct (~63%)
- Gemma 4 26b: 92 correct (~37%)

Table Nodes

```
[sec-2/sec-0/table-0] Selected peptides tested for their potential to inhibit PHF6 fibrillization using ThT assay.
        Columns: Name | Sequence | Selection | Inhibition of PHF6 fibrillization
        [sec-2/sec-0/table-0/row-0] APT | d -aptllrlhslga | d -vqivyk | +
        [sec-2/sec-0/table-0/row-1] KNT | d-kntpqhrklrls | d -vqivyk | +
        [sec-2/sec-0/table-0/row-2] TL28 | l-TTSLQMRLYYPP | l -vqivyk | +
        [sec-2/sec-0/table-0/row-3] TD28 | d-ttslqmrlyypp | d -vqivyk | +
        [sec-2/sec-0/table-0/row-4] TD28rev | d-ppyylrmqlstt | d -vqivyk | +
        [sec-2/sec-0/table-0/row-5] Positive Control | d-tlkivw [ 19 ] | d -vqivyk | +
        [sec-2/sec-0/table-0/row-6] TD2 | d-stfdpltlhktr | d -vqivyk | -
        [sec-2/sec-0/table-0/row-7] TD12 | d-dknhvspvqiyk | d -vqivyk | +/-
        [sec-2/sec-0/table-0/row-8] TL8 | l-sshnvnggtwgs | l -vqivyk | +/-
        [sec-2/sec-0/table-0/row-9] TL18 | l-mnlsngevynak | l -vqivyk | -
```

```
<table frame="hsides" rules="groups">
    <colgroup span="1">
        <col align="left" valign="middle" span="1" />
        <col align="left" valign="middle" span="1" />
        <col align="left" valign="middle" span="1" />
        <col align="left" valign="middle" span="1" />
    </colgroup>
    <thead>
        <tr>
            <th align="left" rowspan="1" colspan="1">Name</th>
            <th align="left" rowspan="1" colspan="1">Sequence</th>
            <th align="left" rowspan="1" colspan="1">Selection</th>
            <th align="left" rowspan="1" colspan="1">Inhibition of PHF6
                fibrillization</th>
        </tr>
    </thead>
    <tbody>
        <tr>
            <td align="left" rowspan="1" colspan="1">
                <bold>APT</bold>
            </td>
            <td align="left" rowspan="1" colspan="1">
                <monospace>
                    <sc>d</sc>-aptllrlhslga
                </monospace>
            </td>
            <td align="left" rowspan="1" colspan="1">
                <sc>d</sc>-vqivyk
            </td>
            <td align="left" rowspan="1" colspan="1">+</td>
        </tr>
        <tr>
            <td align="left" rowspan="1" colspan="1">
                <bold>KNT</bold>
            </td>
            <td align="left" rowspan="1" colspan="1">
                <monospace>d-kntpqhrklrls</monospace>
            </td>
```

Table Nodes (2)

Tested the 38 papers with no/low extraction but classified as relevant (from previous weeks)

Paper_id: 257 / had 1 / got 9 (all from a single table)

Paper_id: 958 / had 0 / got 1

Paper_id: 2530 / had 0 / got 9 (all from a single table)

Paper_id: 4936 / had 0 / got 5

Pipeline Usage

- Parameters for individual phases can be added by a YAML file to setup the pipeline

```yaml
log_level: "INFO"
ncbi_batch_size: 100

# List of steps to run in order
steps:
  - search
  - extraction
  - evaluation

search:
  provider: "ncbi"
  db: "pmc"
  query_list: ["peptide self-assembly"]

extraction:
  provider: "google"
  model_name: "gemini-3.5-flash-lite"
  delay_seconds: 4
  kwargs:
    limit: 3
    id: [130, 257, 941, 958, 1074]

evaluation:
  save: True
  kwargs:
    batch_size: 10
```

Question about Logging

Which approach is better?

1) Directory per parts of pipeline (search, extraction etc.) and separate log files with name “{run_identifier}.log” for each phase

or

2) Combined log file per pipeline run with file name as the program start time

```
#path=section_name/run_identifier.log
logs
```

1

```
Start Time: {time.time()}
Pipeline Sections:
- section_name_1 (search, extraction etc.)
...
- section_name_n

section_name_1:
identifier=
logs

...
section_name_n:
identifier=
logs
```

2

Next Steps

- Add support for non-pmc papers (pdf to llm ready string) -> **actually had this TODO few weeks ago but was not a priority at the time**

- Finish logging